

A Stability-Aware Adaptive Deep Learning Framework for Robust Intrusion Detection in Network Security

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Abstract

Network intrusion detection systems (IDS) are required to protect the present-day communication infrastructure against the ever evolving cyberattacks. Nevertheless, most of the available IDS solutions have been struggling with unstable features, multifaceted interaction of features, and high imbalance of classes leading to inaccurate detection of low-frequency attacks. To overcome these drawbacks, the present study introduces a deep learning-based system, SAFARI-IDS (Stability-Aware Feature Refinement and Adaptive Robust Intrusion Detection System) that is aimed at enhancing the accuracy of detection, its robustness, and its computational efficiency. The architecture incorporates four modules such as Stability-Aware Feature Refinement (SAFR) to eliminate the unstable and redundant attributes, Interaction-Driven Representation Learning (IDRL) to learn the nonlinear associations between features, Cost-Sensitive Adaptive Decision Engine (CADE) to improve minority attack detection by adaptive weighting, and Adaptive

Threshold Stabilization (ATS) to dynamically adapt decision boundaries. The model was tested on the NSL-KDD dataset which consisted of 148,517 records of 41 features and five classes being normal, DoS, Probe, R2L, and U2R. Refined features were 28 in number. Through the experimentation, it has been demonstrated that the SAFARI-IDS performs better than a number of deep learning models based on IDS, with a 98.7% accuracy, 0.974 precision, 0.972 recall, and 0.967 Macro-F1. ROC-AUC of 0.993 and PR-AUC of 0.989 and small variance (0.014) with rapid inferencing were also attained with the system.

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Data Availability

The benchmark dataset specified below is used to evaluate the performances (https://www.kaggle.com/datasets/hassan06/nslddd). The data repository is open access.

Abbreviations

IDS: Intrusion Detection Systems

SAFARI-IDS: Stability-Aware Feature Refinement and Adaptive Robust Intrusion Detection System

SAFR: Stability-Aware Feature Refinement

IDRL: Interaction-Driven Representation Learning

CADE: Cost-Sensitive Adaptive Decision Engine

ATS: Adaptive Threshold Stabilization

IoT: Internet of Things

WSNs: Wireless Sensor Networks

CNNs: Convolutional Neural Networks

RNNs: Recurrent Neural Networks

IoHT: Internet of Health Things

IoMT: Internet of Medical Things

DDoS: Distributed Denial-of-Service

IIoT: Industrial Internet of Things

XAI: eXplainable Artificial Intelligence

LLM-AE-MP: Large Language Model + Auto Encoder + Multi Layer Perceptron

CPSS: Cyber-Physical-Social Systems

R2L: Remote-to-Local

U2R: User-to-Root

FPR: False Positive Rate

FNR: False Negative Rate

DoS: Denial of Service

ROC: Receiver Operating Characteristic

PR: Precision Recall

AUC: Area Under Curve

TPR: True Positive Rate

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Contributions

M.Rekha and. Senthilkumar contributed to the study, conception, methodology, design, and manuscript preparation.

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Ethics declarations

Competing interests

The authors declare no competing interests.

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